Anthony Zhang

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http://antoniomacaronio.github.io/personal-website/#/home
https://github.com/antoniomacaronio

Interests

3D Computer Vision, Video Models, Dextrous Manipulation, Humanoid Robotics, Real2Sim2Real

Education

University of California, Berkeley *08/2025 - Present*
Masters in Electrical Engineering and Computer Science
- Accelerated Master's program with advisor Prof. Angjoo Kanazawa

University of California, Berkeley (GPA: 3.8 / 4.0) *08/2020 - 12/2024*
Bachelors in Computer Science
Bachelors in Applied Mathematics

Publications

See Google Scholar for more information. Publications below are in order from most recent to least recent.

[1] Brent Yi, Chung Min Kim, Justin Kerr, Gina Wu, Rebecca Feng, **Anthony Zhang**, Jonas Kulhanek, Hongsuk Choi, Yi Ma, Matthew Tancik, and Angjoo Kanazawa. *"Viser: Imperative, Web-based 3D Visualization in Python."* In *Proceedings of Robotics: Science and Systems (RSS)*, 2026.

[2] Arthur Allshire*, Hongsuk Choi*, Junyi Zhang*, David McAllister*, **Anthony Zhang**, Chung Min Kim, Trevor Darrell, Pieter Abbeel, Jitendra Malik, and Angjoo Kanazawa. *"Visual Imitation Enables Contextual Humanoid Control."* In *7th Annual Conference on Robot Learning (CoRL)*, 2025. Oral presentation, Best Student Paper Award.

[3] Mingxuan Wu*, Huang Huang*, Justin Kerr, Chung Min Kim, **Anthony Zhang**, Brent Yi, and Angjoo Kanazawa. *"Predict-Optimize-Distill: Self-Improving 4D Object Understanding from Videos."* In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

[4] Lea Muller*, Hongsuk Choi*, **Anthony Zhang**, Brent Yi, Jitendra Malik, and Angjoo Kanazawa. *"Reconstructing People, Places, and Cameras."* In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Highlight Award.

Work Experience

Amazon Frontier AI and Robotics
San Francisco, CA
Applied Scientist Intern *09/2025 - Present*
- Developed egocentric view and language conditioned motion diffusion models to serve as high level planners for humanoid controllers trained on Nymeria's 300hr mocap data.
- Reconstructed 3D human + object trajectories from monocular video and trained VLM-contact-guided motion-tracking RL policies to curate physically plausible HOI data.

Berkeley AI Research
Berkeley, CA
Graduate Student Researcher *01/2024 - Present*
- Developer for Nerfstudio, an open-source framework to build 3D reconstruction methods with +10k Github stars.
- Revamped Nerfstudio's dataloading with multiple parallel workers, optimized disk reads, supported prefetching, and removed unnecessary memory allocations to hasten NeRF and Gaussian Splatting train times by over 55% from 54 minutes to 23 minutes.
- Developed multi-camera support for Meta's Aria virtual reality glasses.
- Implemented spherical harmonics for Gaussian Splatting within GLSL vertex shaders and

Three.js using efficient texture buffers and recurrence calculations, enabling complex view-dependent lighting effects (reflections, highlights, indirect lighting, etc.) for model visualization while achieving 30+ FPS with 1M+ gaussians. 10x their testing QPS.
- Published several papers within computer vision and robotics, receiving 20+ citations.

Anatomage — *Software Developer* — *08/2024 - 03/2025*
Santa Clara, CA
- Developed a mesh compression feature in C++ and VTK that preserved texture UV mapping while achieving up to 90% reduction in polygon count, enabling developers to optimize 3D digital anatomical models in a simple UI.
- Engineered an end-to-end QA automation framework with AutoIt to emulate user behavior with system logging.
- Enhanced Windows application reliability by resolving UI state management issues within the MFC framework.

UC Berkeley EECS — *Teaching Assistant* — *06/2022 - 05/2024*
Berkeley, CA
- Created material, graded homework, developed exams, and handled logistics for discrete math, probability theory, data science, and convex optimization classes at UC Berkeley.

Square — *Software Engineer Intern* — *06/2023 - 08/2023*
San Francisco, CA
- Developed a gRPC admin service using Go to deliver real-time debugging information from a low-latency DynamoDB for on-call engineers, resulting in a 50% reduction in overall debugging time for critical CRM issues.
- Migrated an on-premise monolithic load testing service to the cloud with resources provisioned by Terraform.
- Deployed Docker images to distributed K8s infra with AWS Elastic Container Registry.
- Wrote design document to parallelize load tests across K8 pods, allowing benchmarks to 10x their testing QPS.

Collective Health — *Software Engineer Intern* — *06/2022 - 09/2022*
San Mateo, CA
- Developed SQL queries for Postgres relational databases in Databricks and used PySpark to process and clean various health insurance enrollment statistics from the returned tables for over 300,000 members.
- Built REST API endpoints using Spring Boot to calculate various health metrics based off the tables returned by these queries and routed these HTTP requests into a JSON server to be displayed on the frontend.
- Wrote integration and unit tests of 90% coverage with Mockito to robustify the business logic against exceptions.

Awards
- National Science Foundation Graduate Research Fellowship (2026, Accepted)
- Conference on Robotics Learning (CoRL) 2025: Best Student Paper Award
- Conference on Computer Vision and Pattern Recognition (CVPR) 2025: Paper Highlight Award

Skills
- Programming Languages: Python, Java, GoLang, SQL, C, C++, HTML/CSS, JavaScript. Typescript, ROS2
- Libraries and Frameworks: Spring Boot, gRPC, React, Pytorch, Mockito, JUnit, OpenGL, Three.js